

Rupsha Das

dasrupsha2020@gmail.com | linkedin.com/in/rupsha-das-b6b5a8253 | github.com/Rupsha-Das | +91 9073040582

Experience

Full-Stack Intern, Zedblox Hyderabad - August 2025 - July 2026

- Built and deployed an LLM-powered Telemetry Analytics Assistant using OpenAI GPT and Google Gemini for conversational shipment diagnostics, battery health analysis, anomaly detection, and telemetry insights.
- Developed full-stack features for an IoT management platform supporting 1,000+ connected medical devices, including responsive React interfaces, REST APIs, role-based access control, OTP-protected OTA workflows, metadata versioning, and centralized activity logging.
- Engineered backend services using Node.js, Express.js, MongoDB Aggregation Framework, and REST APIs, applying indexed queries, server-side downsampling, and parallel query execution to reduce API payloads by 65%, improve query performance by 92%, and reduce dashboard latency by 40%.
- Redesigned data models, reusable React components, and analytics dashboards to unify telemetry monitoring across the ActiPod and SafePod device ecosystems.
- Optimized reporting and export pipelines through stream-based CSV/PDF generation and reusable backend services, reducing report generation time by 50% and lowering memory usage.

Embedded Systems Intern, IEEE Computational Intelligence Society Kolkata - June 2025 - July 2025

- Built an autonomous edge-AI vehicle on the ESP32-S3 using a quantized YOLOv5n model for real-time object detection and on-device navigation.
- Developed embedded firmware integrating computer vision, motor control, sensor processing, and wireless communication under strict memory and compute constraints.
- Implemented sensor fusion using ultrasonic sensors and IR wheel encoders to improve obstacle avoidance, localization, and motion accuracy.

Projects

SophistAI - Personal Syllabus Navigator - [sophistai.app](#)

- Developed an LLM-powered learning platform that processes syllabus PDFs and converts unstructured academic content into structured knowledge maps, topic hierarchies, prerequisite relationships, and personalized learning paths.
- Built document-understanding and retrieval workflows to extract course concepts and generate actionable study plans using natural language processing and prompt engineering; secured 1st place among 200+ teams at the Diversion 2k25 Hackathon.

Manimate - Agentic AI Education Platform - [github.com/Rupsha-Das/manimate](#)

- Built an agentic AI education platform using Next.js that transforms academic topics into structured lessons, narrated explanations, and high-fidelity Manim videos through automated lesson planning and Python animation-code generation.
- Designed an end-to-end pipeline combining web research, multi-provider LLM generation, code execution, traceback-aware self-correction, Manim rendering, Kokoro TTS, FFMPEG video stitching, and AI-generated quizzes for interactive learning.

EduSphere - Learning Platform - [github.com/Rupsha-Das/Edusphere](#)

- Built a polished learning-platform frontend using Next.js 16, React 19, TypeScript, Tailwind CSS v4, Radix UI, Framer Motion, and Lucide React, with reusable components and responsive layouts across the application.
- Developed a multi-route learning experience featuring a landing page, playlist discovery, learner dashboard, video player, quiz flow, leaderboard, profile page, progress tracking, and an animated AI assistant interface using a reusable glassmorphism UI system.

Education

University of Kalyani, Kalyani - 2022 - 2026

B.Tech in Computer Science and Engineering DGPA: 7.36

Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frontend: React.js, Next.js, Redux, Tailwind CSS, Framer Motion, Radix UI, responsive UI development, reusable component design

Backend: Node.js, Express.js, Django, Flask, REST APIs, authentication, role-based access control, MongoDB, MySQL

AI & ML: LLM application development, prompt engineering, OpenAI, Gemini, Anthropic, Mistral, TensorFlow Lite, computer vision, YOLOv5, anomaly detection, edge AI, retrieval workflows

Cloud & Tools: AWS S3, AWS EC2, Docker, MongoDB Atlas, Git, GitHub, Postman, Vercel, Netlify, Render, Firebase, Figma, FFmpeg, Manim

Platforms: Linux, Windows, Web, ESP32-S3